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Genome Assembly Using De Bruijn Graph

Project Report

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BONAFIDE CERTIFICATE

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DECLARATION

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List of Abbreviations

- DAA: Design and Analysis of Algorithms
- OLC: Overlap Layout Consensus
- GAGE: Genome Assembly Gold-standard Evaluations
- DNA: Deoxyribonucleic acid
- FASTA: Fast Adaptive Shrinkage Threshold Algorithm
- RNA: Ribonucleic acid

Abstract

While genome sequencing technologies have paved a new path for biological research, resolving complete genomes from fragmented sequencing reads remains a significant setback due to varying read lengths, repetitive regions, and sequencing errors. This project presents a graph-based genome assembly pipeline-the De Bruijn Graph assembly pipeline-that is an elegantly efficient method for reconstructing DNA sequences from short reads. Modelling the overlaps of k-length substrings through k-mer, the pipeline simplifies the assembly process while resolving repetitions effectively and error reduction. Other techniques include bubble removal, tip pruning, and graph optimization conducted on the assembly. Long reads for scaffolding using overlap graphs ameliorate the gaps that short reads cannot bridge, leading to more complete genome assemblies. Quality assessment methods including N50, GC content, and alignment accuracy ensure the biological relevance of the results. This project thus extends the offer of a scalable and precise assembly framework for the assembly of genomes in biomedical research, agriculture, and synthetic biology.

Chapter 1

Introduction

Genome assembly is a major part of bioinformatics as it allows us to understand about an organism and its origins, ancestry, mutation or genetic alterations. It is a process in which scattered parts of the gene called reads, are reconstructed to form the original sequence. To check the correctness method such as alignment is used. Methods such as Overlap Layout Consensus (OLC) and DeBruijn Graphs are used for genome assembly, here in this project we focus on using DeBruijn Graphs to find the genome sequence effectively since they are usually faster and require less memory than the other method.

1.1 Literature Survey

1. An Eulerian path approach to DNA fragment assembly [1]:

This paper discusses about the use Eulerian path in DeBruijn for genome assembly, highlighting the drawbacks of using Hamiltonian path and also significantly improving the computational efficiency.

2. Graph analysis of fragmented long-read bacterial genome assemblies [2]:

In this paper the authors discuss about the use of graph-based technologies in long-read fragmented data assembly and analysis.

3. A survey of mapping algorithms [3]:

They shed light on broad set of sequence mapping algorithm. Focuses on long-read sequencing, which is crucial for complex genome assemblies, highlights the evolving algorithms for more accurate mapping.

4. Genome sequence assembly algorithms and misassembly identification methods [4]:

They reviewed various genome sequence assembly methods for identifying any mis-assemblies in genomes. Provides a comprehensive analysis of multiple algorithms.

5. Computational complexity of sequence alignment and genome assembly [5]:

They analyse the computational complexity of different genome alignment, short-read assemble and sequence comparison.

6. GAGE: A critical evaluation of genome assemblies and assembly algorithms [6]:

This study is known as Genome Assembly Gold-standard Evaluations and focuses mainly on evaluating various genome assemblers. It gives the evaluation metrics and benchmarking for different methods and hence proving useful in comparing models and make a framework to asses genome assembly quality.

1.2 Problem statement

Genome assembly presents several challenges, including handling sequencing errors, resolving repetitive DNA regions, and ensuring computational efficiency in large datasets. Traditional approaches, such as overlap graphs, are computationally expensive for large genomes. This project addresses these issues by leveraging De Bruijn Graphs, which provide a more efficient representation of sequencing data by breaking DNA sequences into smaller overlapping fragments (k-mers) and constructing a graph for genome reconstruction.

1.3 Objectives

This primary objective of this project is to implement a efficient model for genome assembly to reduce the time complexity and improve the overall assembling efficiency compared to other models, the project aims to:

- Efficient Genome Assembly: Using De Bruijn graph for genome assembly.
- Error Handling & Repeat Resolution – Apply computational techniques to correct sequencing errors and resolve genome repeats.
- Performance Optimization – Utilize algorithmic improvements to enhance efficiency
- Comparative Analysis – Evaluate assembly quality across different datasets and k-mer sizes.
- Real-world Applications – Apply results to biomedical research, synthetic biology, and agricultural genomics.

Chapter 2 – Background

Genome assembly is one of the pillars of modern bioinformatics, allowing researchers to reconstruct the complex puzzle of an organism's DNA from scattered sequencing information. It is the foundation of important breakthroughs in medicine, agriculture, and other disciplines. The project under discussion makes use of De Bruijn graphs as an essential strategy for genome assembly, recognized for being able to manage large-scale datasets generated by modern sequencing technology. The chapter covers foundation concepts, issues, and methodologies surrounding genome assembly, with focus placed on the use of De Bruijn graphs, paving the way for the technical workflow described in later sections.

2.1 DNA sequencing fundamentals

This process of genome assembly starts with DNA sequencing, revolutionary technology that interprets the arrangement of nucleotide bases—adenine (A), thymine (T), cytosine (C), and guanine (G)—in a DNA molecule. Existing sequencing technologies, which are mostly next-generation sequencing (NGS) based, produce millions of short stretches of DNA, "reads" that are generally in the range of 100 to 300 base pairs. New long-read technologies, including those from Pacific Biosciences (PacBio) and Oxford Nanopore, push read length to the range of kilobases with associated advantages as well as unique challenges.

Short Reads: Short reads are economical and concise, which makes them good for high-throughput applications but unsuccessful when faced with regions in the genome that contain repeats since their reduced length hinders them from

covering such sequences.

Long Reads: Long reads have the ability to bridge repetitive regions and structural variations, increasing contiguity but with higher error rates as well as higher costs.

The final goal of genome assembly is to reassemble these scattered reads into a coherent sequence, either as continuous segments or full chromosomes, an exercise similar to putting together an extensive, scattered manuscript consisting of overlapping and sometimes inexact pages.

2.2 Challenges in Genome Reconstruction

Constructing a genome is full of obstacles that complicate the route to constructing an accurate reconstruction:

- **Repetitive Elements:** Most genomes contain repetitive elements, such as tandem repeats or transposables, that are longer than standard reads, perplexing attempts to infer their appropriate position and frequently producing fragmented results.
- **Sequencing Imperfections:** Reads with base substitution, insertion, or deletion errors will misguide assembly algorithms, inserting errors into the final sequence.
- **Variability in Coverage:** Incomplete or irregular coverage can underrepresent genomic regions, whereas too much noise from errors can mask actual biological signals.
- **Computational Needs:** The massive amount of data, especially for large eukaryotic genomes, is extremely demanding in terms of both processing power and memory, requiring efficient algorithmic implementations.

Problems highlight the necessity of powerful computational methods that are in a position to handle the intricacies present in genomic information.

2.3 Genome Assembly Strategies

There are two graph-based paradigms that are prevalent in genome assembly: Overlap-Layout-Consensus (OLC) and the De Bruijn graph, with an emphasis in this project on the latter.

- **Overlap-Layout-Consensus (OLC):** OLC builds an overlap graph by determining common sequences of reads, with nodes representing reads and edges representing overlaps. The construction of a layout is followed by consensus sequence development. OLC is useful for reads that are long but is less practical for the massive short-read datasets that are characteristic of NGS due to its computational intensity.

- **De Bruijn Graphs:** Breaking reads down to k-mers—fixed-length k subsequence reads—results in graphs where k-mers are nodes, with overlaps of k-1 bases denoting an edge. The assembly is then reduced to a Eulerian path search, which is computationally more efficient compared to the Hamiltonian path sought in OLC.

Such efficiency and flexibility render De Bruijn graphs most suitable for short-read assembly. The use of De Bruijn graphs in this endeavour takes advantage of their ability to handle large datasets quickly as well as to alleviate some of the pitfalls associated with sequencing errors and repeats.

2.4 De Bruijn Graphs: Mechanism and Use

2.4.1 Graph Construction

The method of De Bruijn graph starts with breaking down every read into overlapping k-mers. For example, from the read "ATCGC" with $k=3$, k-mers "ATC," "TCG," and "CGC" are produced. Each unique k-mer is turned into a node, with k-mers with a $k-1$ overlap linked to each other by directed edges (for example, "ATC" to "TCG" with overlap "TC"). The design of the graph succinctly captures all feasible overlaps in the data, making assembly much more efficient.

2.4.2 Sequence Reconstruction through Eulerian Paths

Assembled genome results from determining a Eulerian path in the De Bruijn graph—a route that crosses every edge only once. The path is then interpreted as a linear DNA sequence, taking advantage of the structure of the graph to maximize computational tractability. Ideally, one Eulerian path is sufficient to solve the genome; with actual data, however, come challenges of disjoint graph elements or more than one feasible path caused by errors and repeats, which then need to be refined.

2.4.3 Improving Graph Quality

In order to offset distortions created by sequencing artefacts, various techniques optimize the De Bruijn graph:

- **Bubble Collapsing:** Divergent paths with the same start and end nodes, frequently resulting from mistakes or genetic aberrations, are identified and merged, keeping only the most supported path by k-mer frequency or coverage.
- **Tip Trimming:** Short, unrelated branches—invariably error-generated dead ends—are cut off, increasing graph coherence by removing confusing segments.

- **Frequency Filtering:** K-mers present with unusually low frequency, characteristic of sequencing errors, are removed, increasing the graph's accuracy to the actual genomic order.

Collectively, these refinements improve the continuity and precision of the resulting assembly by overcoming inherent noise in sequencing data.

2.5 Closing the Gaps Using Long-Read Scaffolding

Whereas short reads have an advantage in constructing non-repetitive areas, their weakness in solving repeats means long reads need to be integrated in scaffolding. The process connects contigs to larger scaffolds by building an overlap graph with contigs or long reads as nodes and representing extensive overlaps with edges. Navigation of this graph orders and orients contigs, closing gaps which cannot be bridged by short reads alone, thus increasing completeness in the assembly.

2.6 Assessing Assembly Quality

Reliability and use of a genome assembly depend on strict quality determination using measures such as:

- **N50:** A contig size metric describing the length where half of the entire assembly exists in contigs of this length or greater, which is an indication of the integrity of the assembled structure.
- **GC content:** The ratio of G and C bases, giving insight into compositional completeness or bias in comparison with the expected genomic composition.
- **Reference Alignment:** When there is a reference genome, accuracy and coverage are quantified by aligning to confirm an assembly to a known standard.
- **BUSCO analysis:** This evaluates the occurrence of conserved single-copy genes to provide a biological estimate of completeness.

These assessments guarantee that the assembly is in line with high-quality standards demanded in downstream genome analysis.

2.7 Broader Implications of Genome Assembly

Applications of genome assembly are widespread across various domains:

- **Medical Discovery:** Constructed genomes shed light upon genetic causes of illness, advance studies of pathogens, and direct targeted therapeutic plans.

- **Crop improvement:** Genomic information highlights characteristics such as hardiness or yield to improve food security.
- **Synthetic Biology:** Engineered genomes serve as blueprints for producing designer organisms with specific functionalities, including bioenergy production or remediation.

De Bruijn graph methodology, which combines efficiency with accuracy, is capable of supporting this project in making such contributions to these disciplines as it leverages the power of genomics data to solve important science problems. This chapter established an extensive foundation, clarifying principles, challenges, and techniques central to genome assembly, with special focus placed upon the use of the De Bruijn graph method. It situates the reader in advance of subsequent methodological explanation, grounding the project in the larger bioinformatics context.

Chapter 3

Methodology

This section describes the systematic pipeline constructed for genome sequence assembly, quality evaluation, and visualization. It is capable of handling raw sequencing to produce high-quality genomic scaffolds. The constructed pipeline can deal with short reads, long reads or files containing both efficiently. The approach makes use of diverse graph-based and computational methods, explained in the subsequent subsections.

3.1 Data Input and Classification

The pipeline begins with the consumption of raw sequencing data (consisting of reads) in FASTA format. Sequences are divided according to their length into short reads (less than a user-defined threshold) and long reads (at or above this threshold). This division allows for customized processing, so the pipeline can use different assembly approaches for each type of read. An optional reference genome can be supplied for alignment and quality comparison. Additionally, user can designate k-mer values, minimum overlap length and

3.2 Short-Read Assembly Using De Bruijn Graph Construction

The classified short reads are passed into a sub-pipeline which employs a De Bruijn graph technique is employed for assembly. In this process sequences are broken into overlapping subsequences of a k-mer size defined by user. To find ideal k-mer size for large dataset user can leverage powers of tools like KmerGenie. The frequency of each k-mer is calculated across all short reads to generate a frequency profile which helps identify potential errors and low coverage reads. A directed graph is constructed using graph theory concepts and python library, where each node represents a k-mer and directed edges signify overlaps between consecutive k-mers, weighted by their occurrence frequency. To reduce noise in the graph, a coverage threshold is applied to filter low-frequency k-mers. Contigs are extracted by identifying starting nodes (nodes with no incoming edges) and sink nodes (nodes with no outgoing edges) and computing paths between them using graph traversal algorithms.

3.3 Graph Optimization Techniques

Refining graphs are essential to improve computational efficiency and ensure quality output. Several graph optimization techniques are applied to refine the De Bruijn graph and improve assembly accuracy:

1. **Bubble Removal:** This technique detects and eliminates alternative paths (bubbles) between the same start and end nodes within the graph. In cases of multiple paths, each path is evaluated with the help of weighted path selection algorithm. This algorithm picks the path with the highest average edge weight and length. This ensures retention of the most biologically plausible assembly by eliminating redundant or erroneous branches.
2. **Entry Tip Removal:** Short branches at the beginning of the De Bruijn graph (entry tips) are identified and removed. This process involves

detecting nodes with no precursor nodes and tracing their successors until a branching point or sink is encountered. Paths are evaluated based on their average edge weights and lengths and the least significant paths are pruned to streamline the graph structure.

3. **Out Tip Removal:** Like entry tip removal, short branches at the end of the graph (out tips) are eliminated. Nodes with no successors are traced backward to identify preceding nodes, and paths are assessed using the same weighted criteria as entry tips. This technique ensures a cleaner, more continuous graph by removing dangling ends that may arise from sequencing errors or low-coverage regions.

These optimization techniques collectively enhance the quality of output and reliability of the graph, minimizing the impact of sequencing artifacts and repetitive regions.

3.4 Long-Read Scaffolding Using Overlap Graph Construction

For the classified long-read data, a scaffolding technique based on overlap graphs is utilized. Overlaps between sequences are detected by marking the shared suffix-prefix regions with a minimum overlap length specified to ensure reliability. An overlap graph is constructed, where nodes represent sequences (either contigs or long reads), and edges denote the length of overlaps. Graph traversal algorithms are applied to compute paths through this graph. It merges sequences by favoring edges with the longest overlaps. The resulting scaffolds are sorted by length to favor the most contiguous and complete assemblies.

3.5 Quality Assessment and Metrics Computation

Quality metrics are calculated to evaluate the assembly's completeness, accuracy, and biological relevance. The following techniques are employed:

- **N50 Calculation:** Contig or scaffold lengths are arranged in descending order and the length at which 50% of the total assembly length is covered is calculated, giving a measure of continuity in assembly.
- **G20 Assessment:** It is a binary metric that helps check if any assembled sequence exceeds a minimum length threshold, indicating the presence of sufficiently long contigs or scaffolds.
- **GC Content Analysis:** The percentage of guanine (G) and cytosine (C) bases in each sequence is calculated and averaged across all assemblies to which characterize and helps in identifying composition of sequence.
- **Alignment and Accuracy Evaluation:** If a reference genome is available, sequences are aligned using a mapping tool optimized for genomic data. Metrics such as the percentage of aligned bases (alignment score) and the proportion of correctly aligned bases (accuracy) are derived to assess similarity to the reference.
- **Quality Control Analysis:** Optional tools are used to profile raw data quality and benchmark assembly completeness against conserved gene sets, evaluating the presence of expected genomic features.

To improve the pipeline, there are sections embedded for use of various tools like Minimap (that aligns long DNA/RNA sequences efficiently), BUSCO (evaluates genome assembly completeness using conserved single copy orthologs) and FASTAQC analyzer for read quality assessment.

3.6 Visualization and Reporting

Visualization techniques are used to simplify interpretation of the assembly results. These include:

- **De Bruijn Graph Visualization:** The structure of the De Bruijn graph is visualized using a force-directed layout algorithm, depicting nodes and edges to illustrate connectivity and complexity. (in cases where dataset is large, we can disable feature)
- **Circular Plot Generation:** A circular (Circos-style) plot is created to show the relative sizes and arrangements of the top scaffolds, providing a spatial overview of the assembly.
- **Distribution Plots:** Histograms are generated to visualize metrics such as GC content distribution and quality scores, assisting in the assessment of assembly uniformity and quality.

All visualizations are stored in a separate output directory and a report is produced pdf format- summarizing key metrics, analysis results and visual outputs.

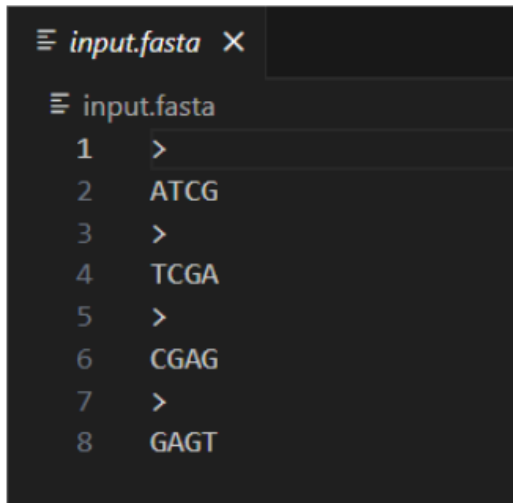
3.7 Output Generation and System Cleanup

The final assembled sequences are exported in FASTA format, with each sequence annotated by its length and formatted for readability. Temporary files and directories, such as intermediate alignment files or quality control outputs, are removed to ensure system efficiency and reproducibility. Error handling and logging mechanisms are built into the pipeline to track progress and deal with possible failure, providing robustness with varied datasets.

Chapter 4

Results

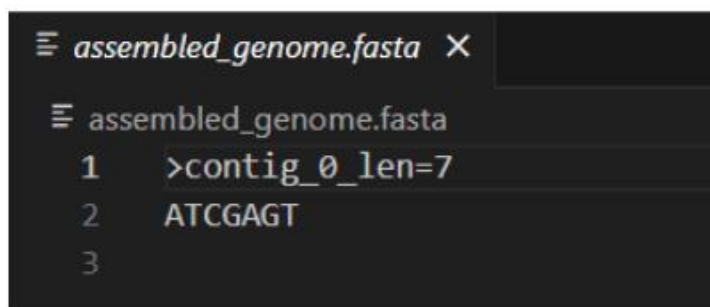
Input file



```
input.fasta X
input.fasta
1 >
2 ATCG
3 >
4 TCGA
5 >
6 CGAG
7 >
8 GAGT
```

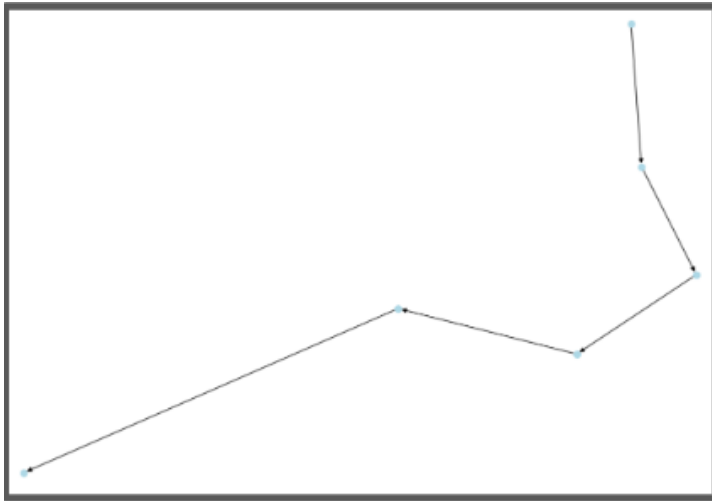
Shows the raw input sequences (reads) in FASTA format used for genome assembly

Result file

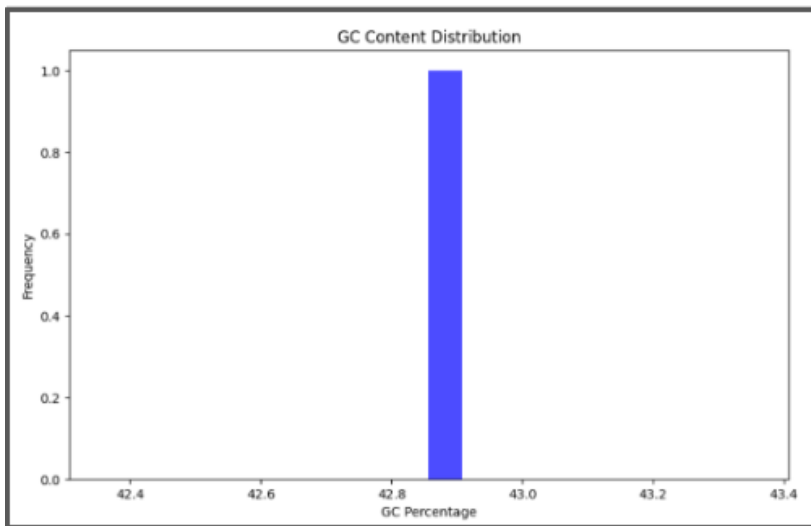


```
assembled_genome.fasta X
assembled_genome.fasta
1 >contig_0_len=7
2 ATCGAGT
3
```

Displays the assembled contigs generated after traversing the De Bruijn graph.



Represents the longer genome sequences (scaffolds) created by connecting contigs using long-read overlaps



Displays the GC content percentage across assembled sequences to evaluate base composition

Assembly Quality Report

Metric	Value
Number of Contigs/Scaffolds	1
Total Length	7 bp
Longest Contig	7 bp
N50	7 bp
G20	Not Performed
GC Content	42.86%
Alignment Score	Not Performed
Accuracy	Not Performed
BUSCO Result	Not Performed
FASTQC Summary	Not Performed

Chapter 5

Conclusion

The exploration demonstrated the usefulness of working out De Bruijn Graphs towards genome assembling, which was fast and accurate. The pipeline outlined was developed to use k-mer composition and graph traversal procedures for constructing contigs out of short reads while coping with typical challenges in next-gen sequencing like errors and repeats. The graph optimization method has improved assembled sequence reliability, with further extension of genome assembly contiguity by long-read scaffolding. Several quality measurements, such as N50, GC content, and alignment accuracy, reflect that the assembly produced is a squeezing. Through these indicators, complete confidence is gained for success from the assembled genome's fitness to the proposed method that has become a very practical and scalable approach regarding genome assembly being useful for a vast number of applications of genomics, health care, and biotechnology.

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